



$$f(x) = 133ax - |x^2 - 10x + 24| > -2$$

1)

$$|x^2 - 10x + 24| < |133ax + 2|$$

$$x^2 - 10x + 24 = 133ax + 2$$

$$x^2 - 10x - 133ax + 22 = 0$$

$$\Delta = (10 + 133a)^2 - 4 \cdot 22 = 0$$

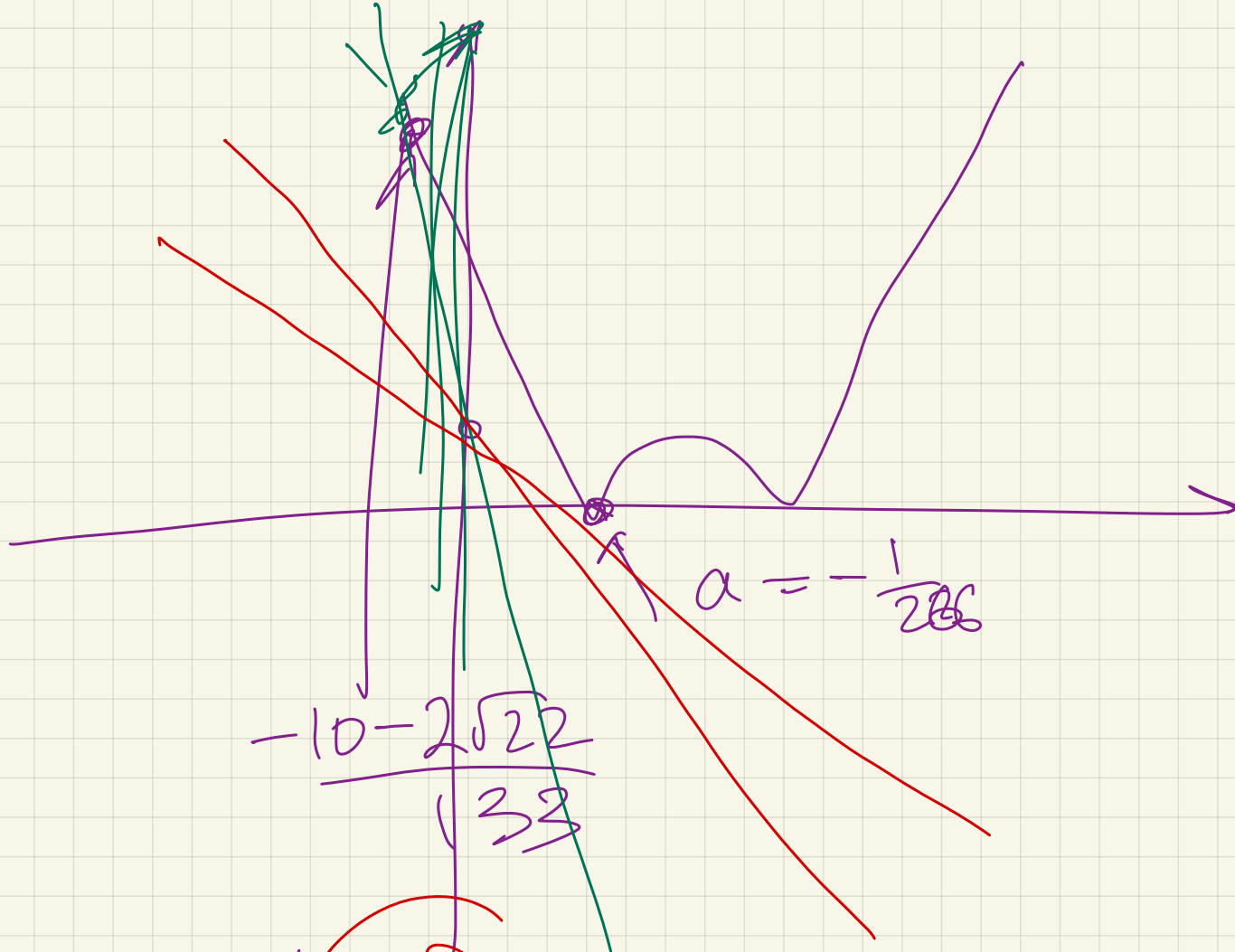
$$10 + 133a = \pm 2\sqrt{22}$$

$$133a = \frac{-10 \pm 2\sqrt{22}}{133}$$

$$y = 133ax + 2$$

$$\frac{-10 + 2\sqrt{22}}{133} = \frac{-1}{133}$$

$$a = \frac{-10 - 2\sqrt{22}}{133}$$



$$a \in \left(-\frac{1}{266}, +\infty \right) \cup \left(-\infty; \frac{-10 - 2\sqrt{22}}{133} \right)$$

$$f(x) = x^2 - 4|x| - ax + a$$

$$[-1; 3]$$

$$\geq -5$$

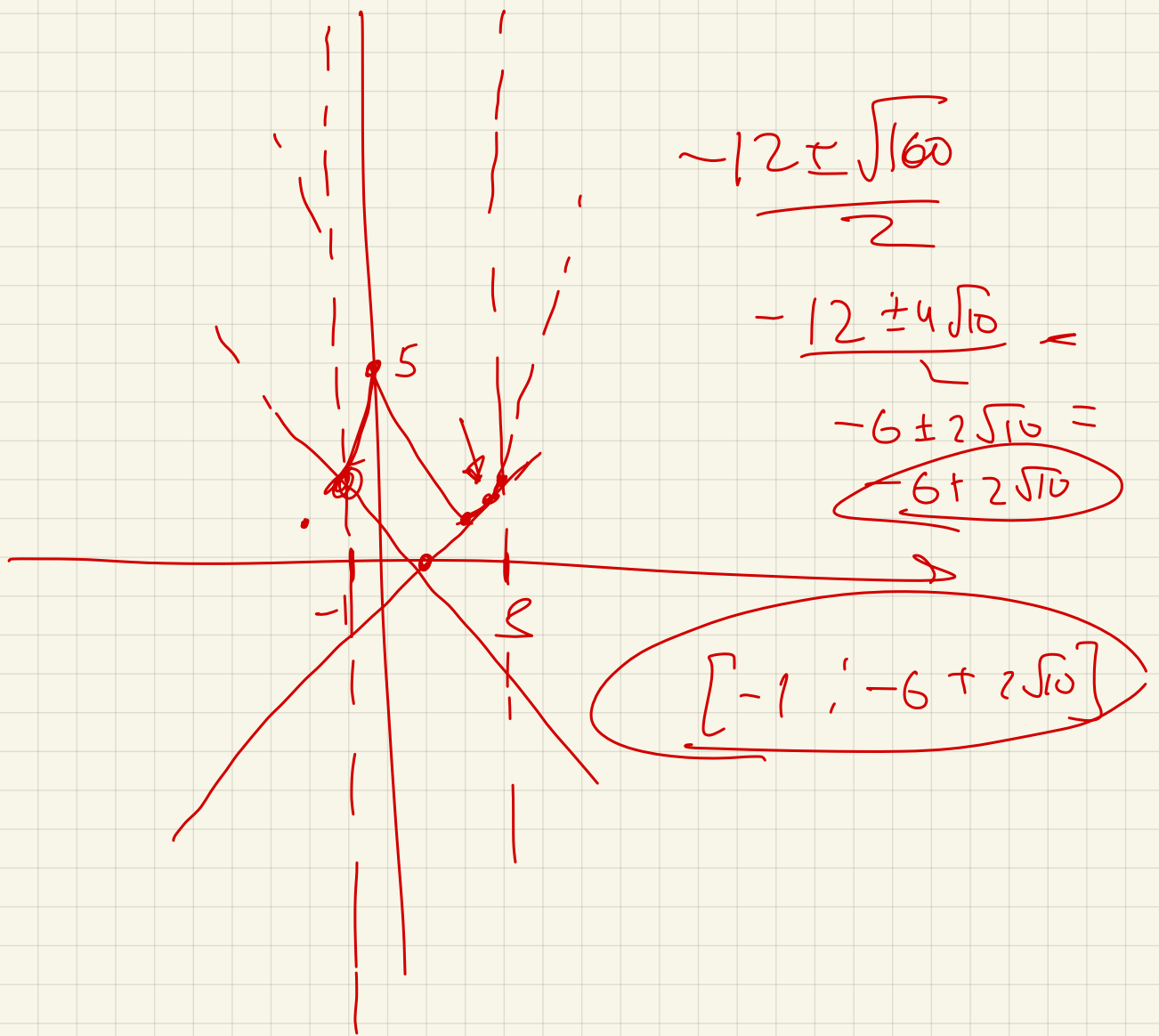
$$x^2 - 4|x| + 5 \geq a(x-1)$$

$$y = x^2 - 4x + 5 \quad x \geq 0$$

$$x_0 = 2 \quad y_0 = 1$$

$$y = x^2 + 4x + 5 \quad x < 0$$

$$x_0 = -2 \quad y_0 = 1$$



$$y = a|x-2| + |a| - 2$$

$$y = \frac{a}{2}$$

S

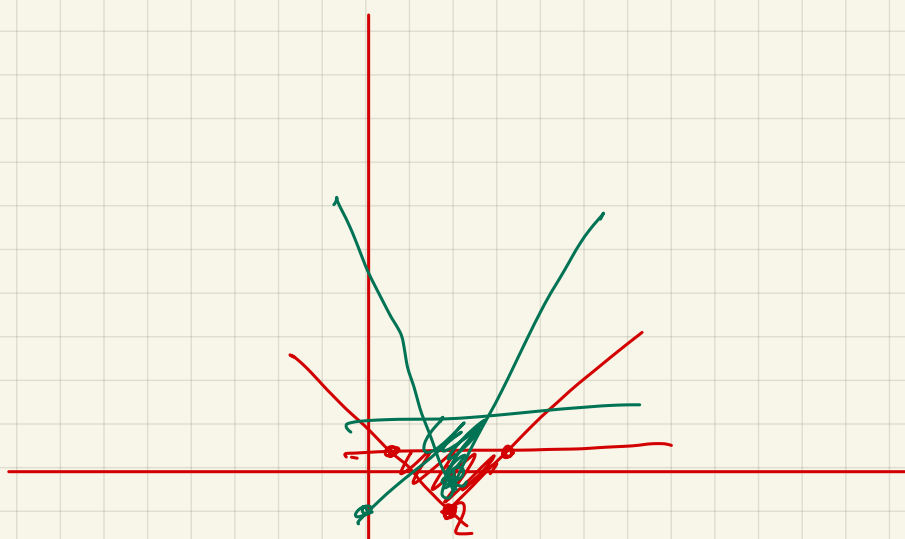
$$a = 1$$

$$y = |x-2| + 1 - 2 = |x-2| - 1$$

$$y = \frac{1}{2}$$

$$a = 2$$

$$2|x-2| + 2 - 2 = 2|x-2|$$



$$a|x| + |a| - 2$$

$$a = 0$$

$$a > 0$$

$$a < 0$$

$$y = -2$$

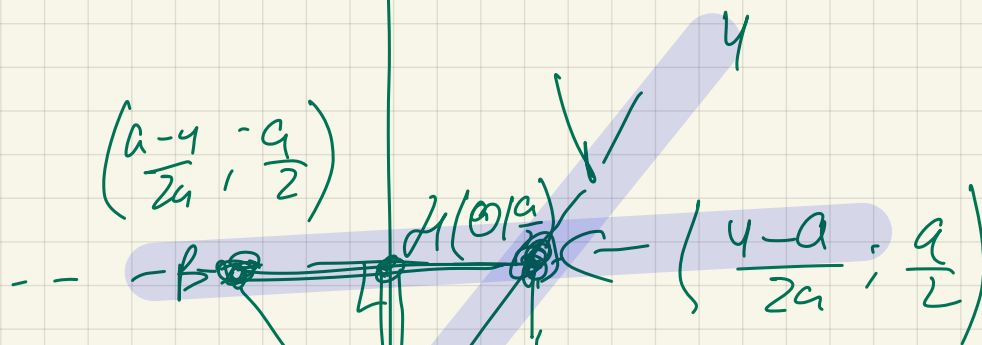
$$y = 0$$

$$y = a|x| + a - 2$$

$$y = \frac{a}{2}$$

$$A(0; a-2)$$

$$\left(\frac{a-y}{2a}, \frac{a}{2}\right)$$



C_i

$$x > 0$$

$$ax + a - 2 = \frac{a}{2}$$

$B:$

$$-ax + a - 2 = \frac{a}{2}$$

$$2ax + 2a - 4 = a$$

$$x = \frac{4-a}{2a}$$

$$y = \frac{a}{2}$$

$$ax = +\frac{a}{2} - 2$$

$$x = \frac{+\frac{a}{2} - 2}{a} =$$

$$\frac{a-4}{2a}$$

$$S = \frac{AH \cdot BC}{2} =$$

$$AH = \sqrt{(0-0)^2 + \left((a-2) - \frac{a}{2}\right)^2} =$$

$$a - 2 - \frac{a}{2} = \sqrt{\left(\frac{a-4}{2}\right)^2} =$$

$$\left| \frac{a-4}{2} \right| = \frac{4-a}{2}$$

$$AH = \frac{4-a}{2} \quad a < 4$$

$$BC = \frac{4-a}{2a} - \left(\frac{a-4}{2a} \right) = \frac{8-2a}{2a} = \frac{4-a}{a}$$

$$f = \frac{1}{2} \left(\frac{4-a}{2} \right) \left(\frac{4-a}{a} \right) \leq \frac{1}{2}$$

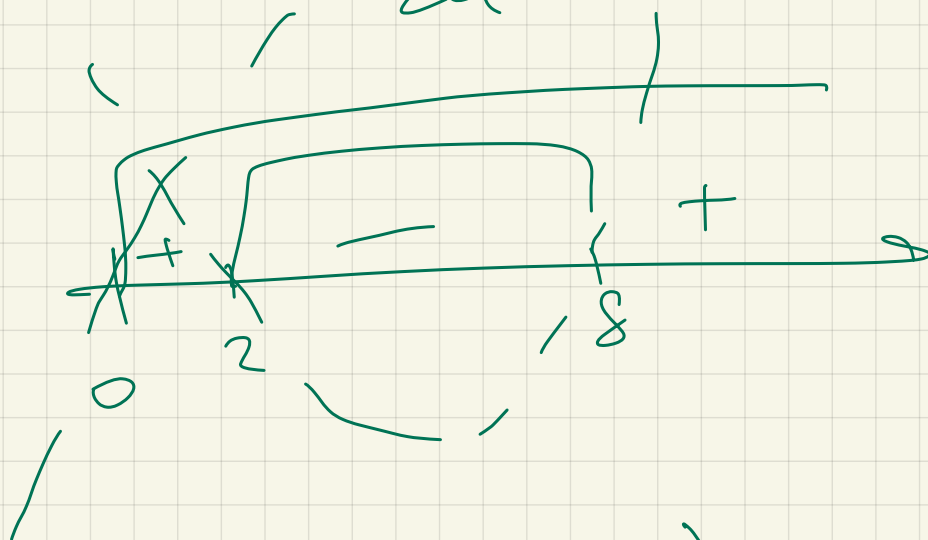
$$\frac{(4-a)^2}{2a} \leq 1$$

$$\frac{16 - 8a + a^2 - 2a}{2a} \leq 0$$

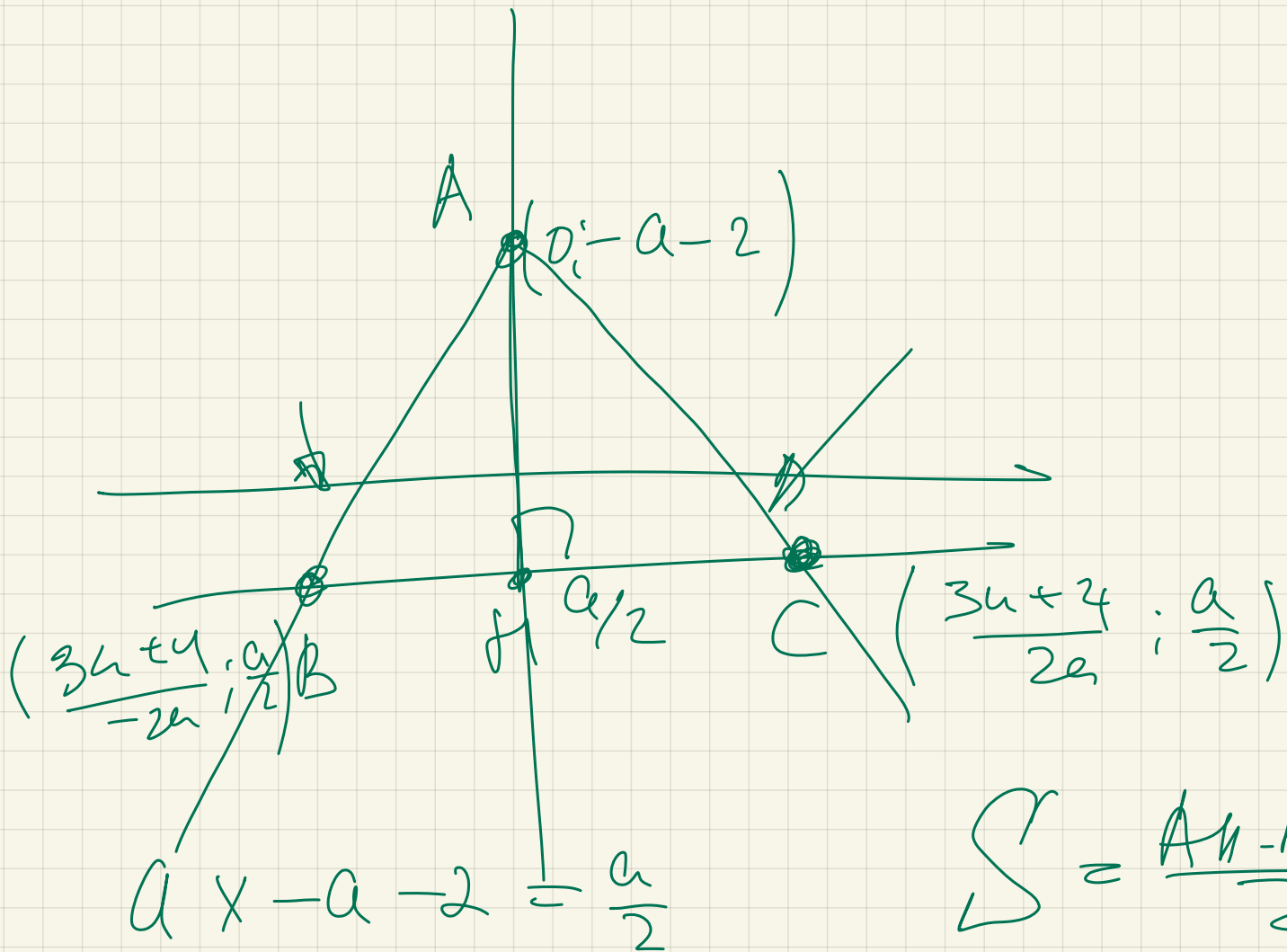
$$\frac{a^2 - 10a + 16}{2a} \leq 0$$

$$\frac{10 \pm \sqrt{100 - 64}}{2}$$

$$\frac{10 \pm 6}{2} \Rightarrow 8, 2$$



$$a \in [2; 8)$$



$$x = \frac{\frac{3a}{2} + 2}{a} = \frac{3a+4}{2a}$$

$$-ax - a - 2 = \frac{a}{2}$$

$$-ax = \frac{a}{2} + a + 2$$

$$x = \frac{\frac{3a}{2} + 2}{-a} = \frac{3a+4}{-2a} =$$

$$\frac{a}{2} - (-a-2)$$

$$\frac{a}{2} + a + 2$$

$$AH = \frac{3a+4}{2}$$

$$\text{BH} \left(\frac{3a+4}{a} \right)$$

$$\begin{cases} \frac{1}{2} \left(\frac{3a+4}{a} \right) \left(\frac{3a+4}{2} \right) \leq \frac{1}{2} \\ a < -\frac{4}{3} \end{cases}$$

$$\begin{aligned} 3a+4 &> 0 \\ a &< 0 \end{aligned}$$

$$\left(\frac{3a+4}{2a} \right)^2 \leq 1$$

$$\frac{9a^2 + 16 + 24a - 2a}{2a} \leq 0$$

$$\frac{9a^2 + 22a + 16}{2a} \leq 0$$

$$a \in \left[-2; -\frac{4}{3}\right) \cup [2; 4)$$

$$f(x) = x^3 - 3ax^2 + 3a^2x - 3|x| + 3$$

2. Skizze $[-2; 3]$

$$f'(x) = 3x^2 - 6ax + 3a^2 - 3 = 0 \quad x \geq 0$$

$[0:3]$ $y_{ca.}$ $z_{wp.}$

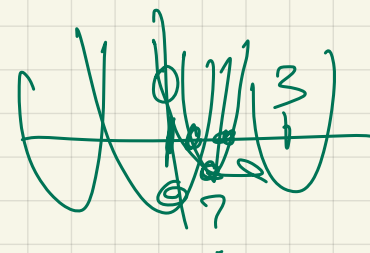
$$x^2 - 2ax + a^2 - 1 = 0$$

$$(x-a)^2 - 1 = 0$$

$$(x-a-1)(x-a+1) = 0$$

$$X = a + 1$$

$$X = a - 1$$



Нес. шаг $b \neq 0$

$$a - 1 > 0$$

$$a + 1 < 3$$

$$a > 1$$

$$a \in (1; 2]$$

$$a < 2$$

Вс. шаг

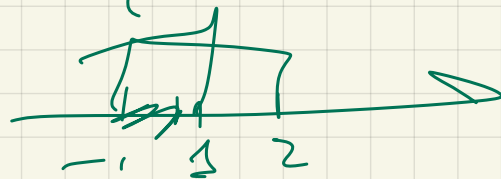
$$a - 1 \leq 0$$

$$0 < a + 1 < 3$$

$\Rightarrow$

$$a \leq 1$$

$$-1 < a < 2$$



$$0 < a - 1 < 3$$

$$a + 1 \geq 3$$

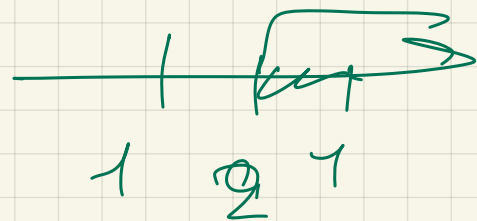
$\Rightarrow$

$$1 < a < 4$$

$$a \geq 2$$

$$a \in (-1; 1]$$

$$\cup [2; 4)$$



$$x^2 - 2ax + a^2 - 1$$

$$a = \pm 1$$

$$x^2 - 2ax$$